

Exercise 3 Report

Objective:

The exercise focuses on building a writer identification system using the ICDAR17 dataset. It involves generating a codebook with k-means, encoding images with VLAD, and classifying them using SVMs.

Dataset & Experimental Setup

- Dataset folders follow the ICDAR2017 Historical Writer Identification local-feature structure.
- Training descriptors: 1182 samples, Test descriptors: 3600 samples.
- Random seed fixed to 42 for reproducibility.
- Dictionary size $K = 100$ using MiniBatchKMeans.
- Approximately 500,000 training descriptors sampled to learn the codebook.
- Runs using `--overwrite` ensure a fair comparison between configurations.

Steps & Implementation:

- Read train/test label files and build absolute paths for descriptor files using the required suffix.
- Randomly sample descriptors from up to 100 training files and collect roughly 500,000 descriptors in total.
- Learn the visual dictionary with MiniBatchKMeans ($K=100$), save it as `mus.pkl.gz` for reuse.
- For each image, load local descriptors and assign each descriptor to its nearest cluster center using BFMATCHER (hard assignment).
- Compute VLAD by summing residuals (descriptor – cluster center) per cluster and concatenating all K residual vectors.
- Apply power normalization (signed square-root) when `--powernorm` is enabled.
- Bonus – GMP: replace sum pooling with Ridge-based generalized max pooling using γ as regularization.
- Train one Exemplar SVM per test encoding with LinearSVC ($C=1000$, `class_weight='balanced'`).
- Use the 2-normalized SVM weight vector as the refined descriptor for evaluation.

Results:

Experiment	Top-1	mAP
VLAD (sum pooling)	0.8058	0.6175
VLAD + Exemplar SVM	0.8756	0.7312
PowerNorm VLAD	0.8197	0.6302
PowerNorm VLAD + Exemplar SVM	0.8850	0.7534
PowerNorm + GMP ($\gamma=1.0$)	0.8197	0.6302
PowerNorm + GMP ($\gamma=1.0$) + E-SVM	0.8858	0.7538
PowerNorm + GMP ($\gamma=10$)	0.8206	0.6304
PowerNorm + GMP ($\gamma=10$) + E-SVM	0.8858	0.7535

Comparative Analysis:

- Base VLAD achieves Top-1 = 0.8058 and mAP = 0.6175.
- Applying Exemplar SVM improves performance to Top-1 = 0.8756 and mAP = 0.7312 (Top-1 +6.97 pp, mAP +11.38).
- Power normalization alone improves over base VLAD: Top-1 = 0.8197, mAP = 0.6302 (Top-1 +1.39 pp, mAP +1.28).
- PowerNorm + Exemplar SVM further increases mAP to 0.7534, about +2.21 over VLAD + E-SVM.
- GMP with $\gamma=1.0$ shows a small gain over standard power-normalized pooling before E-SVM (mAP 0.6302 vs 0.6302).
- Best overall result is with PowerNorm + GMP ($\gamma=1.0$) + E-SVM: Top-1 = 0.8858, mAP = 0.7538.
- $\gamma=1.0$ and $\gamma=10$ produce almost identical performance after E-SVM, suggesting stability across these settings

Conclusion

- Exemplar SVM provides the largest improvement in both Top-1 and mAP.
- Power normalization also yields a consistent boost over raw VLAD when recomputed.
- GMP provides a modest additional gain, with $\gamma=1.0$ performing marginally best in these runs.

Result Evidence:

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Solution.py --powernorm --overwrite
#train: 1182
Codebook loaded from disk.
compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [01:17:00:00, 46.26it/s]
VLAD encodings for training computed
evaluate
Top-1 accuracy: 0.8197222222222222 - mAP: 0.6302290781952572
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 1182/1182 [00:25:00:00, 46.96it/s]
VLAD encodings for training computed.
esvm computation
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [15:51:00:00, 3.78it/s]
Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.885 - mAP: 0.7533722599540954

(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Solution.py --powernorm --gmp --gamma 1.0 --overwrite
#train: 1182
Codebook loaded from disk.
compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [04:44:00:00, 12.64it/s]
VLAD encodings for training computed
evaluate
Top-1 accuracy: 0.8197222222222222 - mAP: 0.6302339486999146
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 1182/1182 [01:33:00:00, 12.64it/s]
VLAD encodings for training computed.
esvm computation
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [15:50:00:00, 3.79it/s]
Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8858333333333334 - mAP: 0.7537659758401607

(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Solution.py --powernorm --gmp --gamma 10 --overwrite
#train: 1182
Codebook loaded from disk.
compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [04:42:00:00, 12.75it/s]
VLAD encodings for training computed
evaluate
Top-1 accuracy: 0.8205555555555556 - mAP: 0.6303947417082116
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 1182/1182 [01:32:00:00, 12.79it/s]
VLAD encodings for training computed.
esvm computation
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [15:51:00:00, 3.78it/s]
Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8858333333333334 - mAP: 0.7534618828981063

(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Solution.py --overwrite
#train: 1182
Codebook loaded from disk.
compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [01:18:00:00, 46.08it/s]
VLAD encodings for training computed
evaluate
Top-1 accuracy: 0.8058333333333333 - mAP: 0.617453263489768
Computing VLAD encodings for training data...
100%|██████████████████████████████████████████████████████████████████████████████| 1182/1182 [00:25:00:00, 46.99it/s]
VLAD encodings for training computed.
esvm computation
100%|██████████████████████████████████████████████████████████████████████████████| 3600/3600 [16:00:00:00, 3.75it/s]
Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8755555555555556 - mAP: 0.7312484540004084
```